Discussion 4: Human Intuition and the Traveling Salesman Problem

Discussion Topic:

The traveling salesman problem is the perfect example of the exponentially increasing complexity of some problems that might seem simple to human beings. It asks the following question: "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city and returns to the origin city?"

It is also an example of an "NP-hard" problem. Machine learning techniques excel at this type of problem. Despite its apparent simplicity, why is the traveling salesman problem challenging for human intuition to solve efficiently as the number of cities increases? What cognitive limitations come into play?

My Post:

Hello Class

In computer science, more specifically in computational complexity theory, “NP-hard” problems are at least as difficult as the hardest problems in the Nondeterministic Polynomial time (NP) class of problems; however, they do not necessarily belong to the NP class (National Institute of Standards and Technology, 2021). An NP problem has solutions that can be verified in polynomial time, meaning the time required to check the solutions increases at a manageable rate as the problem size grows. Similarly, an NP-hard problem can have possible solutions that are relatively easy to evaluate; however, for NP-hard problems that require finding the optimal solution, determining which solution is the optimal solution as the problem size grows requires a significant amount of computation, making finding the optimal solution costly and impractical to compute. In other words, as the NP-hard problem becomes larger, the number of possible solutions that need to be evaluated grows so rapidly that finding the best solution becomes computationally costly and impractical.

The Traveling Salesman Problem (TSP) is such a problem. TSP asks for the shortest possible route a salesman can take to visit each city once and return to the city of origin. In TSP, the routes grow factorially, as the number of cities increases: $(n-1)!/2$, n = the number of cities. For example, a TSP with 4 cities has $(n-1)!/2 = 3$ possible routes, while a TSP with 100 cities has 4.67×10^{155} possible routes. Computing the best route from 3 possible routes is computationally cheap and a simple task; on the other hand, computing the best route from 4.67×10^{155} possible routes is computationally costly and very complex.

When humans attempt to cognitively solve problems such as the TSP, they are limited by their working memory, as human short-term memory can only store about 3 to 5 chunks of information at a time (Cowan 2021). When solving the TSP, an individual needs to keep track of all evaluated routes, their accumulated distances, and, when evaluating a new route, which cities have already been visited and which cities still need to be visited. Trying to solve a TSP having a large number of cities by evaluating every possible route is mentally impossible for humans. Instead, humans use heuristics that involve using practical rules or mental shortcuts. For example, an individual may adopt a visual or Euclidean approach to finding an optimal route; they might look at an overall map, how cities connect, the shape of possible routes, such as organizing around the outside boundary of the cities map (MacGregor et al., 2004). However, a visual solution that appears optimal is not necessarily the absolute optimal route.

Machine Learning (ML) and computational search techniques, on the other hand, can be used to solve TSP problems computationally. ML can easily learn patterns from a city location and distance dataset and, based on those patterns, identify a possible optimum route without having to evaluate every possible route. Moreover, when combined with computational search techniques, an ML model performs significantly better than relying on learned patterns alone (François et al., 2019). For example, François et al. (2019)'s research shows that a hybrid approach to TSP, combining ML with Lin-Kernighan heuristic, a local-search heuristic for TSP, reduced the optimality gap from 7.05% to 0.24%. The optimality gap measures how far a proposed optimal solution is from the best possible optimal solution.

$$Optimality\ Gap\ = \frac{\text{Approximate Route Length} - \text{Optimal Route Length}}{Optimal\ Rout\ Lenght} * 100$$

In conclusion, the TSP is an NP-hard problem that is mentally difficult to impossible for humans to solve as its size increases. Nonetheless, humans rely on heuristics that involve using practical rules or mental shortcuts to find a possible optimal route. On the other hand, a hybrid ML and heuristic search approach may find a possible optimal route that is much closer to the absolute optimal solution than human ones, but it still does not guarantee that the route found is the actual absolute optimal route. Ultimately, the TSP is a perfect example of a problem that is easy to understand, but it is extremely difficult to find the optimal solution as the size of the problem increases, even for ML or AI models using heuristic search tools.

-Alex

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